

POLITECNICO
MILANO 1863

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Requirement Analysis and Specification Document

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Chapter 1

Introduction

1.1 Purpose

The purpose of the *Best Bike Paths (BBP)* system is to promote cycling as a sustainable, safe, and conscious means of transportation by providing cyclists with a digital platform for collecting, accessing, and sharing information about cycling routes and activities.
The system addresses the need for reliable and up-to-date data on bike paths, allowing users to track their performance while contributing to the improvement of shared information.
Through active community participation, BBP aims to identify and maintain the safest and most efficient routes, fostering a more informed and widespread cycling mobility.
Ultimately, BBP seeks to enhance the overall cycling experience by integrating multiple functionalities and creating a collaborative, continuously updated information network driven by user contributions.

1.1.1 Goals

G1. **Support cyclists in tracking their activities:** Enable users to record their cycling routes and access detailed statistics such as total distance, average speed, and other performance indicators, enriched, when available, by meteorological data from external services (weather conditions, temperature and wind speed).

G2. **Allow cyclists to contribute information about bike paths:** Provide two modes for entering information related to bike paths:

- **Manual mode:** the user manually inputs the path data and its condition.
- **Automatic mode:** the system automatically collects data via GPS, accelerometer, and gyroscope during the cycling activity, requiring user confirmation before publication.

G3. **Enable the publication of recorded path information:** Allow users to make their recorded path information public, contributing to a shared and valuable database for the entire cycling community.

G4. **Enable route search and visualization:** Allow users to search for and display on a map one or more cycling routes connecting a chosen origin and destination, ranking them based on overall condition and effectiveness in reaching the goal.

G5. **Maintain up-to-date and reliable path information:** Ensure continuous updating of path data through the aggregation and analysis of user feedback, taking into account data freshness, consistency among reports, and user confirmations.

1.2 Scope

The main goal of the *Best Bike Paths (BBP)* system is to provide cyclists with a mobile application that allows them to record their activities and share information about bike routes, while also fostering the creation of an active and collaborative community. The application enables registered users to save their rides, view detailed statistics such as total distance covered or average speed, and contribute to updating the path inventory by reporting the condition of routes or the presence of potential obstacles. The information entered by users can be made public and is then aggregated by the system, which combines data from multiple sources to offer a comprehensive and up-to-date representation of the paths' conditions. Unregistered users can still take advantage of the consultation features by searching for routes between two points and viewing the available options on a map, ranked according to a score that considers both the quality and effectiveness of the routes. The system also interacts with an external meteorological service, which provides weather data to enrich the information related to recorded activities.

1.2.1 World Phenomena

1.2.2 Shared Phenomena

1.3 Definitions, Acronyms, and Abbreviations

1.3.1 Acronyms

- **BBP:** Best Bike Paths
- **DB:** DataBase

1.4 Revision History

1.5 Reference Documents

1.6 Document Structure

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Chapter 2

Overall Description

2.1 Product Overview

2.2 Product Functions

Sign-Up and Login

Sign-Up

The system lets the users register in order to manage and record data associated with each of them. When the user accesses the sign-up page for the first time, he's offered the option to log-in with one of their existing accounts through an OAuth service. This way, his BBP account is linked to another account that he owns, making it easier to remember and access for him. The system also gives the user the possibility of entering his email and a new password and to create an account directly inside BBP. In that case, BBP checks that the password is complex enough and that no other account is associated with the same email. If the checks pass, the system stores the information in the DB and grants the user access to the platform.

Login

If the user already has a BBP account or has previously linked their BBP account to one of the other existing accounts that he has through OAuth, the system grants them access the page. If the user access the page through OAuth, a button redirects them to the external account provider's website, which handles the login process and sends the user back only when he completes the login procedure. If, on the other hand, the user registered directly on BBP, the system first checks whether an account associated with the provided email, and then verifies that the password matches. If thats the case the user is redirected to the website's homepage.

Accessing BBP without accounts

The user can also access BBP without logging in. In this case, the functionalities that are available are more limited. The system lets the user browse between public paths, but all the recording features and the private paths needs the login to work

Activity Recording

Chapter 3

Specific Requirements

Organize this section according to the rules defined in the project description.

Chapter 4

Formal Analysis Using Alloy

Organize this section according to the rules defined in the project description.

Chapter 5

Effort Spent

Data	Alessio Pesavento	Alberto Spezzapria	Luca Suzani
27/10/2025	1h	1h	1h
07/11/2025	2h	2h	2h
08/11/2025	-	1h	-
09/11/2025	1h	-	-
10/11/2025	-	-	1h
11/11/2025	1,5h	1,5h	1,5h
13/11/2025	1,5h	-	-

Table 5.1: Hours of work spent by each author on each day on the RASD document